ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

ProSoft PLX3x Multi-Protocol Gateway line (PLX31 / PLX32 dual-Ethernet) · Component type: **Embedded Device (EDR) - autonomous multi-protocol conversion gateway (Modbus / EtherNet-IP / DNP3)** · Target SL = SL 3

Where to work	Configure with **ProSoft Configuration Builder (PCB)** on a Windows host: add the gateway (**Choose Module Type** dialog - here you **disable unused ports/protocols**, "UNCHECK IF NOT USED"), set each Ethernet port's IP (PLX32 = two ports on **different subnets**), define the Modbus/EtherNet-IP maps, then **Download** to the gateway. The gateway also serves a **web diagnostics page** (HTTP) that can **upgrade firmware**. Native hardening = the **MODE 3 setup jumper** (disables PCB upload/download **and** web-page/firmware access) + port disable + segmentation. There is **no user login** on the gateway. (In-chassis MVI56E/MVI69: same PCB flow + **ProSoft Discovery Service (PDS)** for IP, owned by a Logix controller.)
Ports / paths	**PCB config:** UDP **50000** (discovery) + UDP **3000** (config & online diagnostics). **Web diagnostics/firmware:** HTTP **80** (no login - lock down via MODE 3). **Application protocols:** Modbus TCP **502**; EtherNet/IP **TCP 44818 / UDP 2222**; DNP3 **20000**; BACnet/IP UDP **47808**; encapsulated Modbus TCP **2000** (model-dependent). PLX32 bridges **two Ethernet networks on separate subnets** - that separation is its core function. No HTTPS/TLS on the gateway.
Notes / CVEs	ProSoft PLX3x multi-protocol gateway (PLX31/PLX32). Autonomous conduit - no owning controller, no device user accounts, no comms encryption. Hardening leans on the **MODE 3 jumper** (fw v1.04+), disabling unused ports, dual-subnet segmentation, protecting the PCB **config file (.ppf)** on the PC, and firmware patching. Track **ProSoft security advisories / CISA ICS advisories** for the specific model + firmware.
How to read	Default read-only (PCB read-only project/config review; read-only Modbus/EtherNet-IP enumeration; read-only web diagnostics). Each row = native step-by-step + command-level external-tool validation + pass. Write/exploit tools = isolated env + written authorization. Autonomous gateway: several rows are Compensating / N-A-at-device.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 1.1	Human user identification & auth	The gateway has **no user accounts / no login** - anyone who can reach **PCB (UDP 50000/3000)** or the **web page (HTTP 80)** can read/reconfigure it. Compensate: set the **MODE 3 jumper** (blocks PCB upload/download **and** the web page) after commissioning, restrict the config subnet by segmentation, and limit who has PCB + the config file on authorized engineering hosts. **Test:** from an unauthorized host attempt a PCB connect / browse `http://` -> blocked by segmentation + MODE 3. **Ext. tools:** Physical check (MODE 3 jumper set); `nmap -sU -p 50000,3000` + `curl -i http:///` from an unauthorized segment (expect unreachable).	MODE 3 set + segmentation restrict who can configure; engineering hosts controlled; no anonymous reconfig.	EDR - no device login; compensating.
SR/CR 1.2	Device & software-process I&A;	The bridged protocols are **unauthenticated** (Modbus TCP, EtherNet/IP, DNP3). Constrain machine-to-machine trust by **segmentation + the gateway's two-subnet separation** and, where the protocol supports it, addressing/whitelists (e.g. DNP3 addressing, EtherNet/IP peer restriction). Document each peer on each side. **Test:** from a non-approved host on either subnet, open Modbus/EtherNet-IP to the gateway -> blocked. **Ext. tools:** PCB (per-port peer/map review); `nmap --script modbus-discover -p 502` ; `nmap --script enip-info -p 44818` ; Wireshark `modbus || enip`.	Peers constrained by segmentation + dual-subnet separation; unauthenticated protocols compensated.	No native device auth - Compensating.
SR/CR 1.3	Account management	No accounts on the gateway; manage engineering identities at the OS/AD level on the PCB host (who may run PCB, who holds the **.ppf config file**) with least privilege and no shared logins. **Ext. tools:** Host/AD account review; PCB-host + config-file custody record.	Engineering identities managed on the PCB host (least privilege); config-file custody controlled.	
SR/CR 1.4	Identifier management	Confirm a unique device identity per gateway (unique IP on each of the two subnets, unique protocol node/unit IDs) and that each bridged peer is uniquely identified/documented; no reuse across sites. **Ext. tools:** PCB (Ethernet config + protocol addresses); `nmap --script modbus-discover -p 502`; network inventory.	Unique gateway + peer identifiers on both subnets; no reuse.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 1.5	Authenticator management	The gateway stores **no device credentials**; the only authenticator-like control is the **MODE 3 jumper** (physical) plus OS credentials on the PCB host. Enforce strong host/AD passwords for the engineering workstation; there is no gateway password to set. **Ext. tools:** PCB-host OS/AD password policy; confirm MODE 3 jumper state.	Strong PCB-host credentials; MODE 3 physical control in place; gateway-credential gap documented.	Gateway: no credentials.
SR 1.6	Wireless access management	**N/A.** Wired DIN-rail gateway with no native wireless interface. **Ext. tools:** Confirm no wireless; upstream network governs.	N/A - justification recorded (wired device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A / limited at device.** PLX3x has no device PKI/TLS (HTTP-only web, no HTTPS; plaintext PCB and application protocols). Record N/A; provide confidentiality/integrity via segmentation and TLS at the SCADA layer. **Ext. tools:** Confirm no configurable certificate; Wireshark (web/PCB/protocols = cleartext).	N/A/limited device PKI; compensating segmentation / upstream TLS.	N/A/Partial.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	**N/A at gateway** (no device login). Confirm the PCB-host OS login masks the secret. **Ext. tools:** Visual verification (PCB host).	Authenticators obscured at the PCB host (N/A at gateway).	N/A at device.
SR/CR 1.11	Unsuccessful login attempts	The gateway has **no login / no lockout**. Remove exposure with the MODE 3 jumper + segmentation; enforce lockout at the PCB-host OS/AD; monitor failed protocol connection attempts on both subnets. **Ext. tools:** PCB-host OS/AD lockout policy; OT IDS on both gateway segments.	No gateway login to brute-force (MODE 3 + segmentation); host lockout; attempts monitored.	Gateway: no lockout - Compensating.
SR/CR 1.12	System use notification	No banner on the gateway/web diagnostics; enforce the approved use notice on the PCB host (GPO) and any SCADA HMI (compensating). **Ext. tools:** PCB host `gpresult /h gp.html`; HMI banner review.	Use notice at PCB host/HMI (compensating at gateway).	Compensating.
SR 1.13	Access via untrusted networks	The PLX32 sits on **two subnets** - confirm both sides are trusted/controlled segments, neither faces an untrusted network directly; firewall PCB (50000/3000) + web (80) off both operational subnets; VPN for remote engineering; MODE 3 set. **Test:** from an untrusted VLAN, reach 50000/3000/80 or the bridged 502/44818 -> must fail. **Ext. tools:** `nmap -Pn -sTU -p 50000,3000,80,502,44818,2222` from the untrusted side (expect filtered); firewall review both subnets.	Gateway config + bridged protocols unreachable from untrusted networks; both subnets controlled; MODE 3 set.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 2.1	Authorization enforcement	The gateway has **no role model** - configuration authorization is all-or-nothing via PCB/web reachability. Enforce it with the **MODE 3 jumper** (blocks config download/upgrade), segmentation, and PCB-host access control; scope what the gateway can write by the **Modbus/EtherNet-IP command list** (read-only vs write commands) in PCB. **Test:** with MODE 3 set, attempt a PCB Download -> blocked; review the command list for unintended write commands. **Ext. tools:** PCB (command-list write scope); physical MODE 3 check; PCB-host access review.	Config change gated by MODE 3 + segmentation + host control; write commands scoped in the map.	No device roles - Compensating.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 2.3	Use control for portable/mobile devices	Control the PCB laptop and removable media (OS device-control); protect the **.ppf config file** (holds the full gateway config); only authorized firmware loaded; the gateway is DIN-rail in a cabinet - control physical access. **Ext. tools:** PCB-host removable-media GPO; config-file ACL; physical inspection (cabinet).	Engineering media + config file controlled; gateway physical access controlled.	
SR/EDR 2.4	Mobile code	No general-purpose mobile code; the gateway runs only ProSoft firmware and executes the compiled command map - no downloaded scripts. Disable the web page (MODE 3) so the firmware-upgrade path is not exposed. **Ext. tools:** PCB / web (firmware version); confirm MODE 3 disables the web/upgrade path.	Only trusted vendor firmware runs; upgrade path locked (MODE 3); no untrusted mobile code.	EDR-specific.
SR/CR 2.5	Session lock	**N/A at gateway** (no interactive session). Enforce workstation lock on the PCB host (OS screen lock/GPO). **Ext. tools:** PCB-host inactivity-lock GPO.	PCB host locks on inactivity (N/A at gateway).	Compensating.
SR/CR 2.6	Remote session termination	Confirm the gateway bounds concurrent protocol connections per port (server connection limits) so a client cannot exhaust it, and PCB diagnostic sessions can be closed; idle protocol sockets time out. **Ext. tools:** PCB (per-port connection limits / diagnostics); Wireshark (connection count / teardown).	Protocol connections bounded per port; diagnostic sessions terminable.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	The gateway has **no native security/syslog log**. Capture change attribution at the PCB host (config-file version control + who downloaded) and log protocol connection/write events on **both subnets** with a **Modbus/EtherNet-IP-aware OT IDS**; forward to the SIEM. **Ext. tools:** PCB config-file version control; OT IDS on both segments + SIEM.	Change attribution at the PCB host; bridged-protocol events captured by IDS on both sides.	No native syslog - Compensating.
SR/CR 2.9	Audit storage capacity	Because logging is off-device (IDS / SIEM / PCB version control), confirm those are sized/retained and alert on collection failure so gateway-related events are not silently lost. **Ext. tools:** SIEM/IDS retention + source-silent alert review.	Off-device logs sized/retained; audit-failure alerting defined.	Compensating.
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	The gateway typically has no synced RTC; confirm event correlation relies on the time-synced IDS/SIEM and endpoint controllers/SCADA (NTP) so bridged-event times align. **Ext. tools:** IDS/SIEM + endpoint NTP review; `nmap -sU -p 123 .`	Correlatable time via synced IDS/SIEM/endpoints.	
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: gateway in a **locked cabinet**; set the **MODE 3 jumper** (its purpose is to lock PCB config + the web/firmware path); secure the **serial config/debug port** (PLX31) and the Ethernet ports; disable unused ports/protocols in PCB. **Ext. tools:** Physical inspection (cabinet, MODE 3 jumper, serial port); PCB (disabled unused ports).	Physical/diagnostic interfaces controlled; MODE 3 set; unused ports disabled.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 3.1	Communication integrity	All PLX3x paths - PCB, web, and the bridged Modbus/EtherNet-IP/DNP3 - are **unauthenticated cleartext** (no TLS/HTTPS) - integrity/confidentiality come from **containment**. Keep each side in its controlled segment; the PLX32's two-subnet separation is the boundary; no uncontrolled cleartext crosses to an untrusted network. **Test:** confirm bridged protocols + PCB/web stay within the two controlled subnets. **Ext. tools:** Wireshark `modbus || enip || http` (cleartext, in-controlled-segment only); firewall review both subnets.	Bridged + config comms contained/segmented; two-subnet boundary enforced; no uncontrolled cleartext.	No device crypto - Compensating.
SR/EDR 3.2	Protection from malicious code (component)	Confirm only ProSoft firmware runs. The firmware-upgrade path is the web page - keep it locked (MODE 3) except during controlled maintenance; obtain firmware from ProSoft and verify provenance; track ProSoft/CISA advisories. **Ext. tools:** PCB / web (firmware version); ProSoft download provenance; ProSoft security advisory / CISA review.	Only vendor firmware runs; upgrade path locked; provenance verified; patched.	EDR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: MODE 3 set, unused ports disabled, both subnets segmented, command-list write scope, firmware patched, against the hardening baseline; compare the PCB **.ppf** config to the approved baseline. **Ext. tools:** PCB (config vs baseline .ppf); `nmap -sTU -p 50000,3000,80,502,44818,2222` (only required reachable).	Security settings match the hardening baseline after change.	
SR/CR 3.4	Software & information integrity	Protect the **PCB .ppf config file** (host ACLs, version control, offline copy) - it defines the entire gateway behavior; hash it to detect drift; MODE 3 prevents unauthorized on-gateway reconfiguration. **Ext. tools:** Host ACL on the .ppf; `certutil -hashfile .ppf SHA256` for drift; MODE 3 check.	Config-file integrity protected + change-tracked; unauthorized reconfig prevented (MODE 3).	
SR/CR 3.5	Input validation	Confirm the gateway tolerates malformed **Modbus / EtherNet-IP / DNP3** frames on either side without faulting or corrupting the map. **ISOLATED only:** fuzz each protocol on a bench gateway and watch the scan/port health. **Ext. tools:** Protocol fuzzer (`boofuzz` Modbus/EtherNet-IP) on a bench gateway; Wireshark; watch gateway diagnostics/port LEDs.	No fault or map corruption on malformed input.	Do NOT fuzz an in-service gateway.
SR/CR 3.6	Deterministic output	Verify configured **failure behavior**: on loss of one side, the gateway drives configured defaults/hold-last for the mapped points (Command Error / Delay Preset behavior) so a stale read does not present as live data to the other side. **Ext. tools:** PCB (command error / comm-fail default behavior); simulate one-side loss -> defined output/data state.	Mapped points assume a defined/fail-safe state on one-side comm loss.	
SR/CR 3.7	Error handling	Confirm graceful fault handling: gateway **port status / diagnostics** (via PCB or the web page) report comm faults and error counters without unsafe behavior or info leak. **Ext. tools:** PCB diagnostics (port status/error counters); status-LED review.	Graceful fault handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No authenticated/encrypted session on any interface; integrity is provided by containment. Record N/A + compensating segmentation. **Ext. tools:** Wireshark (confirm in-segment containment; no cross-boundary replay path).	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-device, protect the IDS / SIEM store and the PCB config-file version history; restrict who can read/clear them. **Ext. tools:** SIEM access review; config-file repository ACL.	Audit records protected from unauthorized read/modify/delete.	
EDR 3.10	Support for updates	Update firmware via the ProSoft web page / firmware tool with firmware from ProSoft; temporarily clear MODE 3 only in a controlled maintenance window, verify version/provenance, then re-set MODE 3; keep the prior firmware for rollback. **Ext. tools:** Web/firmware tool (update flow); ProSoft provenance/version; ProSoft advisory.	Controlled updates from a verified source, patched, with rollback; MODE 3 re-set after.	EDR-specific.
EDR 3.11	Physical tamper resistance/detection	Keep the gateway in a locked/sealed cabinet; the **MODE 3 jumper** is the tamper-resistance control for config/firmware; control the serial + Ethernet ports; config drift is visible via PCB compare. **Ext. tools:** Physical inspection (cabinet/seal, MODE 3 jumper); PCB config compare.	Cabinet secured; MODE 3 set; interfaces controlled; drift detectable.	EDR-specific.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
EDR 3.12	Provisioning product-supplier roots of trust	**Partial/N-A at device.** No documented signed-firmware supplier root of trust on the PLX3x. Record the gap; compensate with firmware-provenance discipline (3.10), the MODE 3 upgrade lock, patching, and segmentation. **Ext. tools:** Confirm no signed-boot/RoT; ProSoft provenance process review.	No documented supplier RoT; compensating controls recorded.	Partial/N-A.
EDR 3.13	Provisioning asset-owner roots of trust	**Partial/N-A at device.** No asset-owner certificate provisioning (no PKI/TLS on the gateway). Record the gap; asset-owner trust is enforced at the network layer (firewall/segmentation/VPN). **Ext. tools:** Confirm no device certificate provisioning; segmentation review.	No asset-owner RoT; compensating controls recorded.	Partial/N-A.
EDR 3.14	Integrity of the boot process	**Partial/N-A at device.** No published verified/secure boot on the PLX3x. Record the gap; mitigate via controlled firmware provenance (3.10), the MODE 3 upgrade lock, patching, physical control, and monitoring. **Ext. tools:** Confirm no secure-boot capability; firmware version review.	No documented verified boot; compensating controls recorded.	Partial/N-A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/repurpose, clear the gateway config (download an empty/default project via PCB) and destroy/secure the **.ppf config file** + backups so the maps, IPs, and peer lists are not recoverable; sanitize any removable media. **Ext. tools:** PCB (download default/blank project); config-file + backup sanitize.	Gateway config + maps purged on decommission or reuse.	
SR/CR 4.3	Use of cryptography	**N/A / limited.** No comms crypto on the gateway (no HTTPS/TLS). Rely on segmentation and TLS at the SCADA layer; lock the web page via MODE 3. Where confidentiality is required across the conduit, add an external secure tunnel. **Ext. tools:** Confirm no device crypto; MODE 3 (web locked); Wireshark (no cleartext across untrusted boundary).	No device crypto - compensating TLS/segmentation upstream; web locked.	N/A/Partial.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 5.1	Network segmentation	Primary control. The PLX32 is a **conduit between two protocol networks** - confirm its two Ethernet ports are on **different, controlled subnets** behind firewalls, only the intended Modbus/EtherNet-IP flows cross, and neither side is flat with IT. PCB (50000/3000) + web (80) are restricted; MODE 3 set. **Ext. tools:** `nmap -sTU -p 50000,3000,80,502,44818,2222` on each subnet; firewall/conduit rule review both sides.	Gateway bridges two segmented subnets; only required flows cross; config planes restricted.	
SR 5.2	Zone boundary protection	Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device). **Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Zone boundary protection verified individually.	
SR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Single-purpose protocol gateway with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR 5.4	Application partitioning	Partitioning the two protocol networks is the gateway's core function - confirm the two sides are separated (dual subnets + firewall), only mapped registers/points cross, and unused ports/protocols are disabled in PCB so no extra path exists. **Ext. tools:** PCB (disabled unused ports + map review); `nmap` per subnet.	Two protocol networks partitioned; only mapped points cross; extra paths disabled.	

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 6.1	Audit log accessibility	The gateway has no native syslog, so monitoring is off-device: confirm a passive **Modbus/EtherNet-IP-aware OT IDS** (Nozomi/Dragos) on **both** gateway subnets captures connection/write/exploitation events and forwards to the SIEM; PCB diagnostics + config-file history are exportable. **Ext. tools:** Passive OT IDS on both segments; SIEM; PCB diagnostics/config export.	Gateway events monitored off-device on both sides in near-real-time; config exportable.	No native syslog - Compensating.
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / CVEs
SR/CR 7.1	Denial-of-service protection	Confirm DoS resilience: the gateway is sensitive to protocol floods / connection exhaustion on either port - use firewall rate controls on both subnets and confirm the gateway keeps bridging (or fails safe) under load; disable unused ports to shrink the surface. **Ext. tools:** ISOLATED bench: `hping3 --flood -p 502` / Modbus + EtherNet-IP storm; watch port state + bridge continuity.	Resources bounded; bridge continues (or fails safe) under load; surface minimized.	
SR/CR 7.2	Resource management	Confirm bounded resources: per-port connection limits and command-scan sizing so one side cannot starve the other; disable unused ports/protocols to free resources. **Ext. tools:** PCB (connection limits / command list / scan settings); bench load test.	Resources bounded; no exhaustion-based loss of the bridge.	
SR/CR 7.3	Control system backup	Confirm the PCB **.ppf config** (+ firmware version record) is backed up (versioned, offline) and a restore is tested on a spare gateway for a known-good rebuild. **Ext. tools:** Backup repository; `certutil -hashfile .ppf SHA256`; restore test (spare gateway).	Backups current and recovery validated.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR/CR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden per the ProSoft manual: **set MODE 3**, **disable every unused port/protocol** in PCB (Choose Module Type - UNCHECK IF NOT USED), keep the two subnets segmented, restrict PCB/web reachability, and patch firmware. **Ext. tools:** PCB (MODE 3 + disabled ports); `nmap -sTU -p 50000,3000,80,502,44818,2222` (only required reachable) both subnets.	Least functionality; MODE 3 set; unused ports disabled; config planes restricted; patched.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Inventory the gateway model (PLX31/PLX32 + protocol variant), firmware version, and both-subnet IPs so vulnerabilities track against **ProSoft / CISA** advisories. **Ext. tools:** PCB (module type + firmware); `nmap --script modbus-discover,enip-info`; ProSoft advisory feed.	Accurate model/firmware/subnet inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
ProSoft Configuration Builder (PCB)	Native config, port-disable & diagnostics (all FRs)	Add the gateway (Choose Module Type - disable unused ports, UNCHECK IF NOT USED); set each Ethernet port IP (PLX32 = two subnets); review the Modbus/EtherNet-IP command map (write scope), diagnostics, firmware. Back up the .ppf. Discovery UDP 50000 / config UDP 3000.	Read/authorized project copy.
MODE 3 setup jumper	Native config/firmware lock (1.1, 2.1, 2.13, 3.2, 3.10, 7.6)	The hardware jumper (fw v1.04+) that disables PCB upload/download AND the web page/firmware upgrade - set it after commissioning; clear only in a controlled maintenance window.	Physical control.
Web diagnostics page / curl	Web-exposure & firmware check (1.1, 3.2, 7.6)	`curl -i http:///`; confirm the web page is unreachable when MODE 3 is set / off unused subnets (no unauthenticated diagnostics or firmware upgrade).	Read-only; authorized.
nmap	Port/service exposure, protocol ID & segmentation (1.x, 3.1, 5.x, 7.x)	`nmap -sTU -p 50000,3000,80,502,44818,2222` ; `nmap --script modbus-discover,enip-info`; scan BOTH subnets.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity & conduit boundary (1.2, 3.1, 3.8, 5.x)	Filter `modbus || enip || http`; confirm cleartext stays within the two controlled subnets; inspect the bridge/boundary.	Passive; SPAN/tap only.
Modbus / EtherNet-IP client + fuzzer (isolated)	Peer & input validation (1.2, 3.5)	Read-only queries to confirm which peers/registers answer; `boofuzz` malformed frames to a bench gateway; watch port/scan health.	ISOLATED bench; written authorization.
Config-file (.ppf) hash + version control	Integrity & drift (3.3, 3.4, 7.3)	`certutil -hashfile .ppf SHA256`; keep versioned offline copies; compare to baseline to detect drift.	Protect the file (holds full config).
Passive OT IDS (Modbus/EtherNet-IP-aware)	Monitoring both subnets (2.8, 6.x) - replaces native syslog	Nozomi/Dragos on a SPAN on BOTH gateway segments to capture connection/write/exploitation attempts; forward to the SIEM.	Passive; coordinate with SOC.
ProSoft / CISA advisories	Vuln tracking (3.2, 3.10, 7.8)	Track the specific PLX3x model + firmware against ProSoft security advisories / CISA ICS advisories.	Reference.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.